the remaining two constraints are incorporated via iterative updates to the decoder output distributions at inference time. Our inference-time decoding method is different from the only recent effort by Su et al. (2020) that leverages non-dialog data (forum comments, book snippets) as distant labels to train dialog systems with supervision. Our contributions can be summarized as follows:

- We propose a novel approach to enrich dialog agent personas with relevant backstories, relying only on existing story datasets.
- We propose to use an unsupervised back-propagation based decoding procedure[1] to adapt the relevant stories such that the resulting response is fluent with the dialog history and consistent with the dialog agent persona. Our method works with a model trained just with dialog data i.e. without access to story corpus at training time.
- Our experiments demonstrate that the proposed approach results in much more engaging and specific dialog outputs in a persona-grounded dialog setup. This fills a gap in existing dialog models which often lack the capability to generate responses about specific events and experiences relevant to persona attributes.

2 Unsupervised Persona Enrichment with Background Stories

Given dialog history h and persona C consisting of several (typically 3-5, example shown in Figure 1) attributes, our goal is to construct a dialog response x. Our underlying model is based on the discrete persona attribute choice model from Majumder et al. (2020). To generate a dialog utterance x, we first sample a persona attribute $c \sim p(c|h)$ conditioned on the dialog history h. x is then generated conditioned on the dialog history and the chosen persona attribute. The underlying dialog model's decoder is initialized with a pretrained GPT-2 model, and is fine-tuned on the PersonaChat dataset (Zhang et al., 2018). However, in our current setup, we also have to identify relevant background stories and use them to construct fluent responses at decoding time. Therefore, we propose a different decoding procedure.

To generate a response, we first sample a persona attribute $c \sim p(c|h)$. Next we retrieve stories corresponding to the persona attribute c (Section 2.1). However, the underlying dialog model is trained to generate responses conditioned only on the dialog history and persona. To incorporate the retrieved story in the response, we perform gradient-based inference (Section 2.2), that only assumes a left-to-right language model trained on dialog context and responses, and the story is handled at decoding time in an unsupervised fashion. We refer to the proposed method as **PABST** (Unsupervised **P**erson**A** enrichment with **B**ackground **ST**ories).

2.1 Retrieving Relevant Stories

For a persona attribute c, we aim to identify relevant stories from a story corpus. Toward this goal, we rank the stories using the F1 component of BERT-score (Zhang et al., 2020) based retrieval using the persona attribute c as the query and the highest scoring story is chosen. Note that many of the stories are written in the third person. For use as background stories, we must first transform them to first–person. Following prior work (Brahman and Chaturvedi, 2020), we identify the protagonist of such stories as the most frequently occurring character. Thereafter, we use co-reference resolution (Lee et al., 2017) to identify all words or phrases that refer to the protagonist. Finally, all words or phrases so identified are replaced with suitable first person pronouns (e.g. 'his books' to 'my books').

2.2 Gradient-based Inference

Our underlying dialog model is not trained to condition on a retrieved story, and cannot be directly used to construct a desirable response using s. To tackle this, we consider a decoding strategy which, in addition to fluency with history h, encourages response x to follow two soft constraints: (1) be minimally different from story s, and (2) be consistent with persona c.

First, we generate an initial response based only on the dialog history. Then we perform an iterative procedure which alternates between performing a forward pass on the language model to encourage fluency, and a backward pass which updates the response via back-propagation to respect the two soft constraints. However, x is discrete, and cannot be directly updated using gradients from back-propagation. Instead, we maintain and update a soft representation o of x, where o_i corresponds to the last hidden state representation for the i^{th} token position, i.e., $p(x_i) \sim \mathrm{softmax}(Wo_i/\tau)$, where τ is the temperature parameter, W is the embedding

[1] Code can be found at https://github.com/majumderb/pabst

matrix, and $Wo_i \in \mathcal{R}^V$ (V is the vocabulary size). Our approach is inspired by recent works that use gradient-based decoding for text generation with soft constraints (Dathathri et al., 2020; Qin et al., 2020). Next we describe the backward and forward passes of the iterative procedure.

Backward Pass with Soft Constraints We define the following soft constraints on response x:
(1) **Divergence from story:** We want to encourage x to be *minimally different* from the story s. Following prior work (Qin et al., 2020), we compute a cross entropy loss (denoted by cross-entr henceforth) with story $s = \{s_1, \ldots, s_T\}$ tokens as labels and $Wo_1, \ldots, Wo_T$ as the logits.
(2) **Consistency to persona:** We want x to be *consistent with persona attribute* c. Consider a classifier $q_\phi(o, c)$ which predicts the probability of x (or rather the soft representation o of x) entailing c. The classifier $q_\phi(o, c)$ is a bag-of-words classification head on decoder hidden states o, fine-tuned on the Dialogue-NLI dataset (Welleck et al., 2019) to predict whether pairs of persona attributes and responses are entailed or not. The objective to maximize can be written as:

$$\mathcal{L}(c, s; o) = \lambda_c \log q_\phi(o, c) - \lambda_d \, \text{cross-entr}(s, Wo)$$

where λ_c and λ_d are hyper-parameters. We update o through back-propagation by computing the gradient $\nabla_o \mathcal{L}(c, s; o)$, while keeping the model parameters constant. Let the resulting o after the gradient-based updates be denoted by o^b.

Forward Pass to Encourage Fluency Next we perform a forward pass of the underlying dialog model, with the goal of regularizing the hidden states towards the unmodified language model values. On computing the forward pass at the j^{th} token, we mix the final hidden states o_j^f from the forward pass with o_j^b computed in the backward pass, via weighted addition to get the resulting $o_j = \gamma \times o_j^f + (1 - \gamma) \times o_j^b$, where $\gamma \in (0, 1)$ is a hyperparameter. The resulting o_j is used for computing the logits at the next time step $j + 1$.

We initialize the output response by performing greedy decoding from the underlying dialog model, conditioned on the dialog history and persona attribute. Then we iteratively update o by alternate backward and forward passes. We sample the final response $x \sim \text{softmax}(Wo/\tau)$. In practice, we found that 5 iterations are sufficient to generate good quality outputs.

Method	Training	Decoding	D-1	D-2	ENTR
W/o Story Data					
TRANSFERO	PERSONA	Nucleus	0.05	0.11	1.21
DISCCHOICE	PERSONA	Nucleus	0.15	0.25	1.25
DISCCHOICE	CS-KB	Nucleus	0.87	1.07	2.04
With Story Data					
DISCCHOICE	PSEUDO	Nucleus	0.91	2.45	2.89
DISCCHOICE	MULTITASK	Nucleus	0.99	2.54	2.71
DISCCHOICE	PERSONA	RETRIEVAL	2.56	9.67	3.86
PABST (Ours)	PERSONA	Grad. Inf.	1.56	3.57	3.21

Table 1: Diversity metrics on the PersonaChat test set. D-1/2 is the % of distinct uni- and bi-grams. ENTR is the geometric mean of n-gram entropy. Grad. Inf. is the unsupervised gradient-based decoding as opposed to Nucleus sampling (Holtzman et al., 2020).

3 Experiments

We evaluate methods in terms of their capability to generate diverse, fluent and engaging responses. Hyperparameters are noted in Appendix §A.

Datasets We experiment with the PersonaChat dialog dataset (Zhang et al., 2018) consisting of 131,438 utterances for training, 15,602 for validation, and 15,024 for testing. For stories, we use the training split of the ROCStories dataset (Mostafazadeh et al., 2016), that consists of 78,529 stories, each typically of 4 to 5 sentences.

Baselines We consider two broad groups of models as baselines: **(1)** ***Without access to story corpus***: We use finetuned GPT2 (**TRANSFERO**) on PersonaChat, and the discrete persona attribute choice model (**DISCCHOICE**) from Majumder et al. (2020). We also consider a version of DISCCHOICE which enriches personas with inferences from a commonsense knowledge base (**CS-KB**). **(2)** ***Baselines using story corpus***: To allow DISCCHOICE models to generate story-like responses, we adapt an alternative training regime (**PSEUDO**) from (Su et al., 2020), where we randomly replace some of the target dialog responses with retrieved stories—treating them as pseudo labels. Finally, we also consider a **MULTITASK** training setup from (Su et al., 2020), wherein the decoder is trained on PersonaChat as well as with a language modeling objective on ROCStories. We additionally consider a **RETRIEVAL** baseline that uses the retrieved story verbatim as the dialog response.

3.1 Automatic Evaluation

We hypothesize that that the proposed approach to leverage external non-dialog data can increase the diversity of the generated responses. Following